

Lie Theory: Chapter 1 Homework

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1. Let $\mathbb{R}_\vee^3$ be the Lie algebra defined in Section 1.2. Prove that if $x \cdot y$ denotes the dot product of vectors $x, y \in \mathbb{R}^3$, then

$$x \wedge (y \wedge z) = (x \cdot z)y - (x \cdot y)z \quad \forall x, y \in \mathbb{R}^3.$$

Proof. Observe that

$$\begin{aligned} x \wedge (y \wedge z) &= (x_1, x_2, x_3) \wedge (y_2 z_3 - y_3 z_2, y_3 z_1 - y_1 z_3, y_1 z_2 - y_2 z_1) \\ &= (x_2 y_1 z_2 - x_2 y_2 z_1 - x_3 y_3 z_1 + x_3 y_1 z_3, \\ &\quad x_3 y_2 z_3 - x_3 y_3 z_2 - x_1 y_1 z_2 + x_1 y_2 z_1, \\ &\quad x_1 y_3 z_1 - x_1 y_1 z_3 - x_2 y_2 z_3 + x_2 y_3 z_2) \\ &= (x_2 y_1 z_2 - x_2 y_2 z_1 - x_3 y_3 z_1 + x_3 y_1 z_3, \\ &\quad x_3 y_2 z_3 - x_3 y_3 z_2 - x_1 y_1 z_2 + x_1 y_2 z_1, \\ &\quad x_1 y_3 z_1 - x_1 y_1 z_3 - x_2 y_2 z_3 + x_2 y_3 z_2) \end{aligned}$$

and

$$\begin{aligned} &= (x \cdot z)y - (x \cdot y)z \\ &= (x_1 z_1 + x_2 z_2 + x_3 z_3)(y_1, y_2, y_3) - (x_1 y_1 + x_2 y_2 + x_3 y_3)(z_1, z_2, z_3) \\ &= (x_1 z_1 y_1 + x_2 z_2 y_1 + x_3 z_3 y_1, x_1 z_1 y_2 + x_2 z_2 y_2 + x_3 z_3 y_2, x_1 z_1 y_3 + x_2 z_2 y_3 + x_3 z_3 y_3) \\ &\quad - (x_1 y_1 z_1 + x_2 y_2 z_1 + x_3 y_3 z_1, x_1 y_1 z_2 + x_2 y_2 z_2 + x_3 y_3 z_2, x_1 y_1 z_3 + x_2 y_2 z_3 + x_3 y_3 z_3) \\ &= (x_2 y_1 z_2 - x_2 y_2 z_1 - x_3 y_3 z_1 + x_3 y_1 z_3, \\ &\quad x_3 y_2 z_3 - x_3 y_3 z_2 - x_1 y_1 z_2 + x_1 y_2 z_1, \\ &\quad x_1 y_3 z_1 - x_1 y_1 z_3 - x_2 y_2 z_3 + x_2 y_3 z_2), \end{aligned}$$

where the the last line comes after commuting elements in the terms with same index across all three variables (i.e. $x_1 z_1 y_1$ becomes $x_1 y_1 z_1$) since $\mathbb{R}$ is commutative and then cancelling them out.

We can see that both expressions reduce to the same final expression, so they are equal. $\square$

2. Check that

$$[e_{ij}, e_{kl}] = \delta_{jk} e_{il} - \delta_{il} e_{kj},$$

where δ is the Kronecker delta, defined by $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ otherwise.

Proof. Recall that $[e_{ij}, e_{kl}] = e_{ij}e_{kl} - e_{kl}e_{ij}$. The first term will have zeroes in all but the i^{th} row and zeroes in all but the l^{th} column since the only nonzero row in e_{ij} is the i^{th} one and the only nonzero column in e_{kl} is the l^{th} one. Thus the only possibly-nonzero entry is the one in the i^{th} row and l^{th} column. This will be 1 if and only if $j = k$ (and 0 otherwise) since we will then have the 1 from e_{ij} multiplied by the 1 in e_{kl} and zeroes everywhere else when doing the matrix multiplication. Thus we obtain the first term $\delta_{jk}e_{il}$ of our desired result.

The only other term to deal with is the second one, for which we use the same argumentation on $-e_{kl}e_{ij}$ to conclude that we subtract a 1 from the entry at the j^{th} column of the k^{th} row if and only if $i = l$, and a zero otherwise. This gives us our second term of $-\delta_{il}e_{kj}$, and we are done. $\square$

3. Prove that the set e_{ij} for $i \neq j$ together with the set $e_{ii} - e_{i+1,i+1}$ for $1 \leq i < n$ forms a basis for $\mathfrak{sl}(n, F)$.

Proof. Given a matrix $A \in \mathfrak{sl}(n, F)$, we want to show that we can write A uniquely as a linear combination of the proposed basis elements. The $e_{ii} - e_{i+1,i+1}$ elements have zeroes everywhere outside the diagonal, and the nonzero elements of e_{ij} and e_{kl} do not coincide whenever $i \neq j$ and $k \neq l$. Thus, when $i \neq j$, A_{ij} is the only coefficient of e_{ij} that works in the linear combination, so these “non-diagonal” basis elements have uniquely determined coefficients.

Turning our focus to the “diagonal” basis elements of the form $e_{ii} - e_{i+1,i+1}$, we see that when $i = 1$, there is only one possible coefficient: $A_{1,1}$. This in turn determines the value of $(e_{1,1} - e_{2,2})_{2,2}$ to be $-A_{1,1}$, so since the only other proposed basis element which has a nonzero $(2, 2)$ entry is $e_{2,2} - e_{3,3}$, its coefficient must be $A_{2,2} + A_{1,1}$. Continuing in this manner through $i = n - 1$, we see that the coefficient of $e_{i,i} - e_{i+1,i+1}$ must be $\sum_{j=1}^i A_{j,j}$.

The last thing to do is to check that this linear combination gives us the correct A . As noted during the above investigation, the values are correct by construction for the non-diagonal entries of A and also for the diagonal entries from $A_{1,1}$ through $A_{n-1,n-1}$. For $A_{n,n}$, we need the fact that $\text{tr}(A) = 0$, which tells us that $A_{n,n} = -\sum_{i=1}^{n-1} A_{i,i}$. The (n, n) entry of our linear combination will by construction be $-\sum_{i=1}^{n-1} A_{i,i} = A_{n,n}$, so A is indeed a unique linear combination of the proposed basis elements, and thus they do form a basis. $\square$

4. Prove that an ideal of a Lie algebra is always a subalgebra.

Proof. Let I be an ideal of a Lie algebra L . Then I is a linear subspace of L by virtue of being an ideal, so we just need to show that $[x, y] \in I$ whenever $x, y \in I$. This is quite trivial, since I being an ideal guarantees that for any $x \in L$ and $y \in I$, $[x, y] \in I$, and surely $x \in L$ if $x \in I \subset L$, so there is nothing more to show. $\square$

5. Consider the adjoint homomorphism map defined (on page 4) for all $x, y \in L$ by

$$(\text{ad } x)(y) := [x, y].$$

- (a) Prove that ad is linear.
- (b) Prove that ad is a homomorphism.
- (c) Prove that the kernel of ad is the center of L .

Proof.

(a) For $a, b \in F$ and $x, y, z \in L$,

$$\begin{aligned} (\text{ad}(ax + bz)(y)) &= [ax + bz, y] \\ &= a[x, y] + b[z, y] \quad (\text{by bilinearity of the bracket}) \\ &= a(\text{ad } x)(y) + b(\text{ad } z)(y). \end{aligned}$$

(b) We have already shown that ad is linear, so we just need to show that $\text{ad}[x, y] = [\text{ad } x, \text{ad } y]$. Indeed, for any $x, y, z \in L$

$$\begin{aligned} (\text{ad}[x, y])(z) &= [[x, y], z] \\ &= -[z, [x, y]] \\ &= [x, [y, z]] + [y, [z, x]] \\ &= [x, [y, z]] - [y, [x, z]] \\ &= (\text{ad } x)([y, z]) - (\text{ad } y)([x, z]) \\ &= (\text{ad } x)((\text{ad } y)(z)) - (\text{ad } y)((\text{ad } x)(z)) \\ &= ((\text{ad } x) \circ (\text{ad } y))(z) - ((\text{ad } y) \circ (\text{ad } x))(z) \\ &= [\text{ad } x, \text{ad } y](z). \end{aligned}$$

(c) The center of L is defined as the set of elements x of L that satisfy $[x, y] = 0$ for any $y \in L$. If $x \in L$ is in the kernel of ad , then $\text{ad } x$ is the zero map, so by definition $0 = (\text{ad } x)(y) = [x, y]$ for all $y \in L$. Thus elements of the kernel of ad are in the center, and just as easily we see that if x is in the center of L , $[x, y] = 0$ for all $y \in L$, so $\text{ad } x$ is the zero map. By mutual inclusion then, the kernel of ad is the center of L . □

6. Prove that $\text{Der}(A)$ is a vector subspace of $\mathfrak{gl}(A)$.

Proof. By definition, derivations are linear, so $\text{Der}(A) \subset \mathfrak{gl}(A)$. We need to show, then, that given an algebra A over a field F along with derivations $D, E \in \text{Der}(A)$ and $c \in F$, $D + E \in \text{Der}(A)$ and $cD \in \text{Der}(A)$. Indeed, for any $a, b \in A$, we have

$$\begin{aligned} (D + E)(ab) &= D(ab) + E(ab) \\ &= aD(b) + D(a)b + aE(b) + E(a)b \\ &= aD(b) + aE(b) + D(a)b + E(a)b \\ &= a(D(b) + E(b)) + (D(a) + E(a))b \\ &= a(D + E)(b) + (D + E)(a)b \end{aligned}$$

and

$$\begin{aligned} (cD)(ab) &= c(D(ab)) \\ &= c(aD(b) + D(a)b) \\ &= c(aD(b)) + c(D(a)b) \\ &= a(cD(b)) + (cD(a))b \quad (\text{by bilinearity of algebra multiplication}) \\ &= a(cD)(b) + (cD)(a)b, \end{aligned}$$

so both satisfy the Leibniz rule and are thus derivations as desired. □

7. Let L_1 and L_2 be two abelian Lie algebras. Show that L_1 and L_2 are isomorphic if and only if they have the same dimension.

Proof. First, if L_1 and L_2 are isomorphic Lie algebras, then there is a bijective linear map between them, so they are isomorphic as vector spaces, which forces them to have the same dimension.

Now suppose L_1 and L_2 both have dimension n . Then there exists an isomorphism of vector spaces $f : L_1 \rightarrow L_2$. It is bijective and linear by definition, so we just need to check that the bracket is preserved. Indeed, for any $x, y \in L_1$,

$$\begin{aligned} [f(x), f(y)] &= 0 && \text{(since } L_2 \text{ is abelian)} \\ &= f(0) && \text{(since } f \text{ is a homomorphism of vector spaces)} \\ &= f([x, y]) && \text{(since } L_1 \text{ is abelian)} \end{aligned}$$

as desired. □

8. Find the structure constants of $\mathfrak{sl}(2, F)$ with respect to the basis given by the matrices

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Proof. We have

$$\begin{aligned} [e, f] &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= h, \end{aligned}$$

$$\begin{aligned} [e, h] &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & -2 \\ 0 & 0 \end{pmatrix} \\ &= -2e, \end{aligned}$$

and

$$\begin{aligned}
[f, h] &= \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix} \\
&= 2f.
\end{aligned}$$

Adopting the e, f, h notation in place of the usual indexed notation, we have the structure constants

$$\begin{array}{lll}
a_{ef}^e = 0 & a_{ef}^f = 0 & a_{ef}^h = 1 \\
a_{eh}^e = -2 & a_{eh}^f = 0 & a_{eh}^h = 0 \\
a_{fh}^e = 0 & a_{fh}^f = 2 & a_{fh}^h = 0
\end{array}$$

□

9. Prove that $\mathfrak{sl}(2, \mathbb{C})$ has no non-trivial ideals.

Proof. Let I be a non-zero ideal. We want to show that the basis elements $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, and $h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ are in I . If they are, then the closure of I under linear combinations implies that their span (namely, L) is a subset of I , so $I = L$. We will make use of the results from problem 8 that $[e, f] = h$ and $[f, h] = 2f$.

Let $x = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \in I$. We first calculate

$$\begin{aligned}
[e, x] &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & -a \end{pmatrix} - \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} c & -a \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & a \\ 0 & c \end{pmatrix} \\
&= \begin{pmatrix} c & -2a \\ 0 & -c \end{pmatrix} \\
&= ch - 2ae \in I \text{ by absorption.}
\end{aligned}$$

Since $ch - 2ae \in I$, we can bracket e with it to get

$$\begin{aligned}
[e, ch - 2ae] &= [e, ch] - [e, 2ae] \\
&= c[e, h] - 2a[e, e] \\
&= c[e, h] \\
&= c \left(\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right) \\
&= c \left(\begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right) \\
&= c \begin{pmatrix} 0 & -2 \\ 0 & 0 \end{pmatrix} \\
&= -2ce \in I,
\end{aligned}$$

so $e \in I$. Now we can use absorption again to get $[e, f] = h \in I$. One more application of absorption gives $[f, h] = 2f \in I \implies f \in I$, so we are done. $\square$

10. Let S be an $n \times n$ matrix with entries in a field F . Define

$$\mathfrak{gl}_S(n, F) = \{x \in \mathfrak{gl}(n, F) : x^T S = -Sx\}.$$

(a) Show that $\mathfrak{gl}_S(n, F)$ is a Lie subalgebra of $\mathfrak{gl}(n, F)$.

(b) Find $\mathfrak{gl}_S(2, \mathbb{R})$ if $S = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

Proof. (a) Let $x, y \in \mathfrak{gl}_S(n, F)$ and $a, b \in F$. First we show that $\mathfrak{gl}_S(n, F)$ is a linear subspace, making use of the linearity of the transpose itself:

$$\begin{aligned}
(ax + by)^T S &= (ax^T + by^T)S \\
&= ax^T S + by^T S \\
&= -aSx - bSy \\
&= -Sax - Sby \\
&= -S(ax + by).
\end{aligned}$$

Now we show closure under the bracket:

$$\begin{aligned}
[x, y]^T S &= (xy - yx)^T S \\
&= ((xy)^T - (yx)^T)S \\
&= (y^T x^T - x^T y^T)S \\
&= y^T x^T S - x^T y^T S \\
&= -y^T Sx + x^T Sy \\
&= Syx - Sxy \\
&= S(yx - xy) \\
&= S[y, x] \\
&= -S[x, y].
\end{aligned}$$

(b) For an arbitrary $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathfrak{gl}(2, \mathbb{R})$, we have

$$A^T S = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^T \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & a \\ 0 & b \end{pmatrix}$$

and

$$-SA = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -c & -d \\ 0 & 0 \end{pmatrix}.$$

This tells us that if we have $A^T S = -SA$, setting corresponding entries equal to each other gives $c = 0$, $b = 0$, and $a = -d$. Thus $A = \begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix}$ and we have

$$\mathfrak{gl}_S(2, \mathbb{R}) = \left\{ \begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix} : a \in \mathbb{R} \right\}.$$

□

11. Let A be an algebra and let $\delta : A \rightarrow A$ be a derivation. Prove that δ satisfies the Leibniz Rule

$$\delta^n(xy) = \sum_{r=0}^n \binom{n}{r} \delta^r(x) \delta^{n-r}(y) \quad \forall x, y \in A.$$

Proof. For $n = 1$, the Leibniz Rule simplifies to

$$\delta(xy) = x\delta^1(y) + \delta(x)y,$$

which is already the defining property of a derivation.

Now assume we have proven the theorem through $n - 1$. Then

$$\begin{aligned}
\delta^n(xy) &= \delta(\delta^{n-1}(xy)) \\
&= \delta\left(\sum_{r=0}^{n-1} \binom{n-1}{r} \delta^r(x) \delta^{n-r-1}(y)\right) \\
&= \sum_{r=0}^{n-1} \binom{n-1}{r} \delta(\delta^r(x) \delta^{n-r-1}(y)) \\
&= \sum_{r=0}^{n-1} \left(\binom{n-1}{r} \delta^r(x) \delta^{n-r}(y) + \binom{n-1}{r} \delta^{r+1}(x) \delta^{n-r-1}(y) \right) \\
&= \sum_{r=0}^{n-1} \binom{n-1}{r} \delta^r(x) \delta^{n-r}(y) + \sum_{r=0}^{n-1} \binom{n-1}{r} \delta^{r+1}(x) \delta^{n-r-1}(y) \\
&= \dots \text{I'm not at all sure how to bridge this gap! I've tried lots of things.} \\
&= \left(\sum_{r=0}^{n-1} \binom{n-1}{r} \frac{n}{n-r} \delta^r(x) \delta^{n-r}(y) \right) + \delta^n(x)y \\
&= \left(\sum_{r=0}^{n-1} \binom{n}{r} \delta^r(x) \delta^{n-r}(y) \right) + \delta^n(x)y \\
&= \left(\sum_{r=0}^{n-1} \binom{n}{r} \delta^r(x) \delta^{n-r}(y) \right) + \binom{n}{n} \delta^n(x) \delta^{n-n}(y) \\
&= \sum_{r=0}^n \binom{n}{r} \delta^r(x) \delta^{n-r}(y)
\end{aligned}$$

□